

Job Application Tracker Using Node.js and MongoDB

Project Description

The Job Application Tracker is a backend-based application developed using Node.js, Express.js, and MongoDB to help users manage and track their job and internship applications. The system provides a centralized platform for storing application details such as company name, job role, location, salary, application status, and application date. Since the project is designed as a REST API service, it does not include a frontend interface and is operated entirely through API requests using Postman and console-based outputs.

Objective

The primary objective of this project is to demonstrate practical implementation of MongoDB database operations, RESTful API development, data analytics using aggregation pipelines, API documentation through Swagger UI, and backend application architecture using Node.js and Express.js.

Working of the System

The application follows a client-server architecture where requests are sent through Postman to the Express.js server. The server processes the requests and interacts with MongoDB using Mongoose ODM. When a user submits a new job application, the application details are validated and stored in the MongoDB database. Users can retrieve, update, delete, search, and filter application records through dedicated REST API endpoints.

The system also provides analytical insights using MongoDB Aggregation Framework. These analytics include total applications submitted, application status distribution, company-wise application statistics, average expected salary, and monthly application trends. Seed data functionality is implemented to automatically populate the database with sample records for testing and demonstration purposes.

Swagger UI is integrated into the application to provide interactive API documentation, allowing users to explore and test available endpoints directly from the browser. All operations are performed without a graphical user interface, making the project lightweight, efficient, and focused on backend development concepts.

Technologies Used

- Node.js
- Express.js
- MongoDB
- Mongoose
- Swagger UI
- Postman
- JavaScript
- REST API Architecture

Key Features

- Create, Read, Update, and Delete Job Applications
- Search Applications by Company Name
- Filter Applications by Status
- MongoDB Aggregation-Based Analytics
- Seed Data Generation
- Swagger API Documentation
- Environment Variable Configuration
- Console and API-Based Operation

Expected Outcome

The completed system provides a fully functional backend application capable of managing job application records and generating meaningful analytics. The project demonstrates database design, REST API development, MongoDB integration, aggregation operations, API documentation, and backend software engineering practices without requiring any frontend implementation.